

LEC9 - The Copernican Revolution

Learning Goals

□ Explain the observations and theories that led European Renaissance astronomers to reject the geo-centric model and adopt the helio-centric model of the Solar System

□ Introduce Keplers Laws (Will revisit next class from the lens of planetary orbits)

Renaissance Era: A period of time around 1500-1600CE

- Mona Lisa painted
- William Shakespeare

* You don't need to memorize this

Epicycles: A smaller orbit an object takes as it moves along a much larger orbit

Stellar Parallax:

As Earth orbits Sun, nearby stars should appear to shift positions relative to background stars

* Parallax was measured from 2 different points in Earth's orbit

* While parallax exists, it is too small for naked eye

(> 0.75" [arcseconds] for the CLOSEST star)

• Many models of the Solar System have existed. The earliest were geo-centric, but slowly helio-centric models became favored

• Took place around **renaissance era**

• Views shifted from **Earth at center** of the Solar System to the **Sun at the center**
 geo-centric helio-centric

Note:

Geo → Earth

Helio → Sun

Centric → Centered

Western Astronomy before Copernicus:

Observations:

- Stars, Sun, & Moon rise and set everyday
- It doesn't feel like the Earth is moving
- The stars don't seem to ever change
- Things fall to Earth (gravity)

Assumptions:

- The heavens are perfect & incorruptible
- Circles are perfect

Conclusions:

- Earth is the center of the Universe
- Everything orbits around Earth in perfect circles

Motion of the Planets (Retrograde Motion):

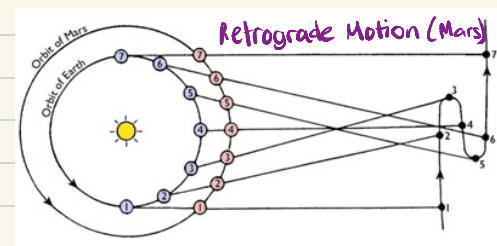
- Planets seemed to move relative to background stars, since they orbit within the Solar System
- At certain times, planets appeared to move "backwards" relative to the background

Ptolemy's model for the orbit of Mars (circa 100AD):

- Explained in a geocentric model
- Accepted in Western Astronomy for 1500 years

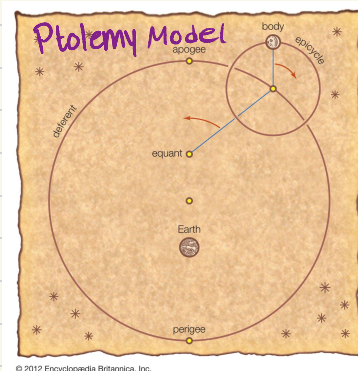
Epicycles:

- Stars orbit Earth in circles
- Planets move in "circles within circles"



Heliocentric Model (Copernicus circa 1500):

- Sun is at center
- Planets (including Earth) orbit around the Sun in perfect circles
- The Moon orbits around the Earth
- Retrograde motion is caused by faster rotation of inner planets

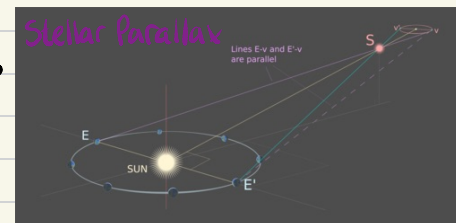


Tycho Brahe (1546-1601):



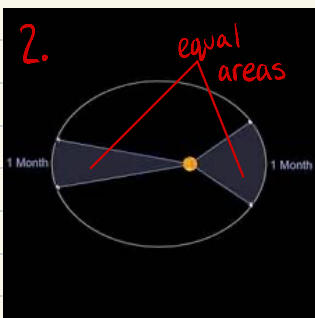
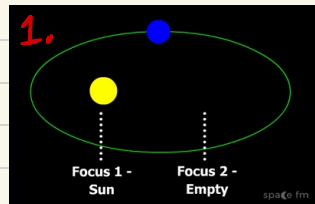
- Danish Nobleman
- Observational Astronomer
- led observatory in making of precise motions of planets & stars
- Naked eye observations (No telescopes)
- Searched for **stellar parallax**
 → Failed to measure it

Earth's position is fixed (geocentrism) **OR** The stars are extremely distant



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Ellipse: A circular shape made of two focal point and an eccentricity from 0-1, where 0 is a circle, and 1 is essentially a line



Galilean Moons: The 4 moons Galileo Galilei discovered

- Io
- Europa
- Ganymede
- Callisto

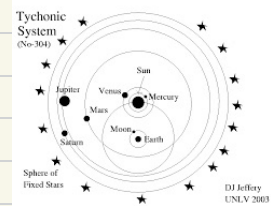
* You don't need to memorize these moon names, only what they are

• Proposed Tychonic model (Hybrid of geo & helio centric models)

Johannes Kepler (1570-1630):



- Middle Class
- Theoretical Astronomer
- Worked with Tycho Brahe's observations and found a fix for the Copernican Model
- Was able to explain Tycho's observations with 3 laws



Keplers Laws:

1. **Keplers First Law:**

The orbit of each planet around the Sun is an **ellipse** with the Sun at one focus

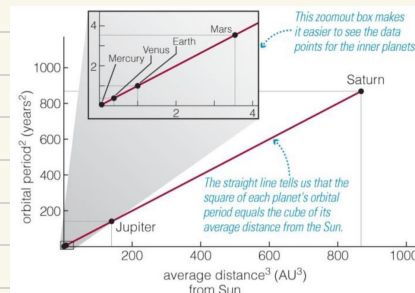
2. **Keplers Second Law:**

As a planet moves around the Sun in its orbit, it sweeps out equal areas in equal times
→ Planets travel faster when closer to the Sun, and slower when further away

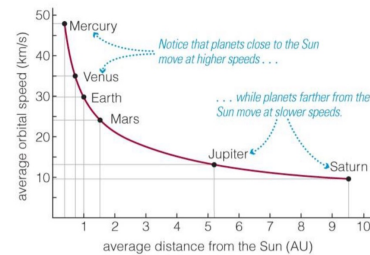
3. **Keplers Third Law:**

More distant planets orbit the Sun at slower average speeds, obeying the relationship:

$$\text{orbital period (years)}^2 = \text{average distance from Sun (AU)}^3$$



a This graph shows that Kepler's third law ($p^2 = a^3$) holds true; the graph shows only the planets known in Kepler's time.



b This graph, based on Kepler's third law and modern values of planetary distances, shows that more distant planets orbit the Sun more slowly.

Galileo Galilei (1564-1642):



- Observed night sky through telescope
- Discovered 4 moons orbiting Jupiter (**Galilean Moons**)
→ Therefore, Earth can't be at the center of the Universe, since these moons orbited another body
- Discovered that Venus has "phases" just like the Moon

The Heliocentric model of the Solar System was preferred in the end because it...

1. Explained retrograde motion
2. Explained phases of Venus
3. Makes extremely accurate predictions about the positions of the planets

- Johannes Kepler created 3 laws for planets that explained the observations of Tycho Brahe
- Galileo Galilei used a telescope to find 4 moons orbiting Jupiter, proving the Solar System is not geo-centric
- Helio-centric model of the Solar System became favored because it explained various phenomenon in the Solar System